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XGUARDIAN: Towards Generalized, Explainable and More Effective Server-side Anti-cheat in First-Person Shooter Games

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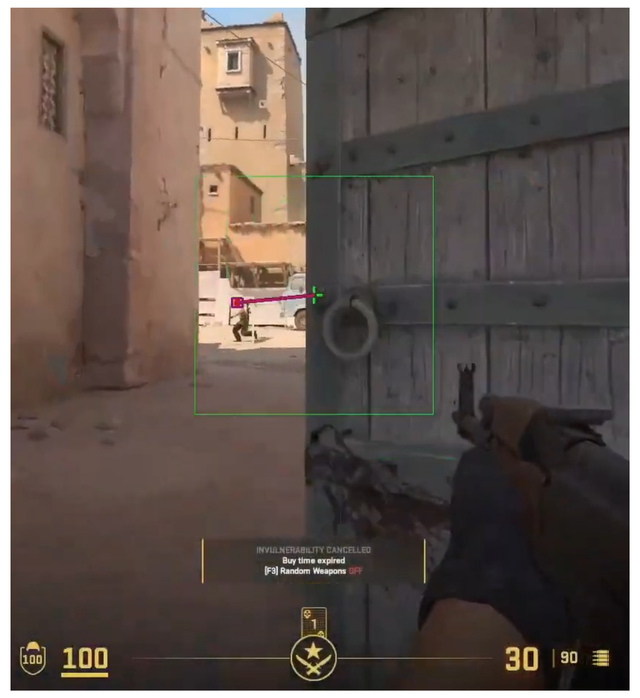
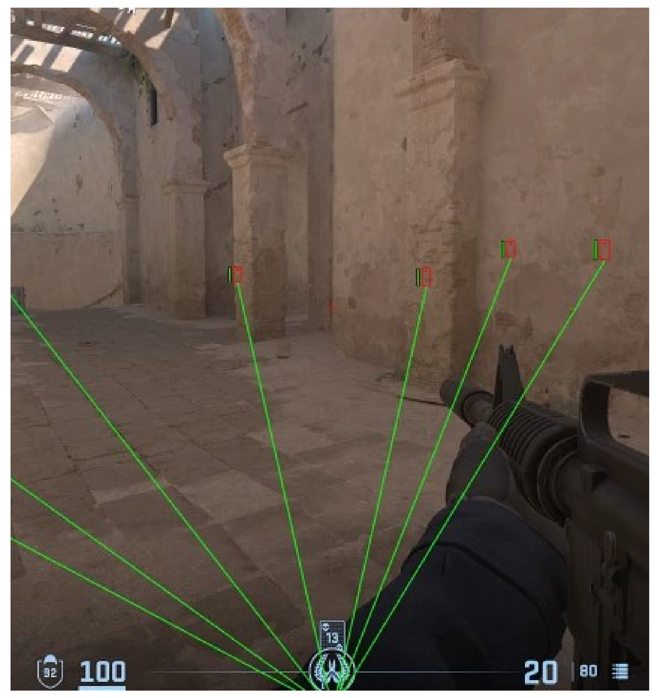
ARTIFACT EVALUATED USENIX AVAILABLE | ARTIFACT EVALUATED USENIX FUNCTIONAL | ARTIFACT EVALUATED USENIX REPRODUCED

To Detect

The most prevalent cheat in FPS: **aim-assist cheats** with real-world anti-cheat scenarios.

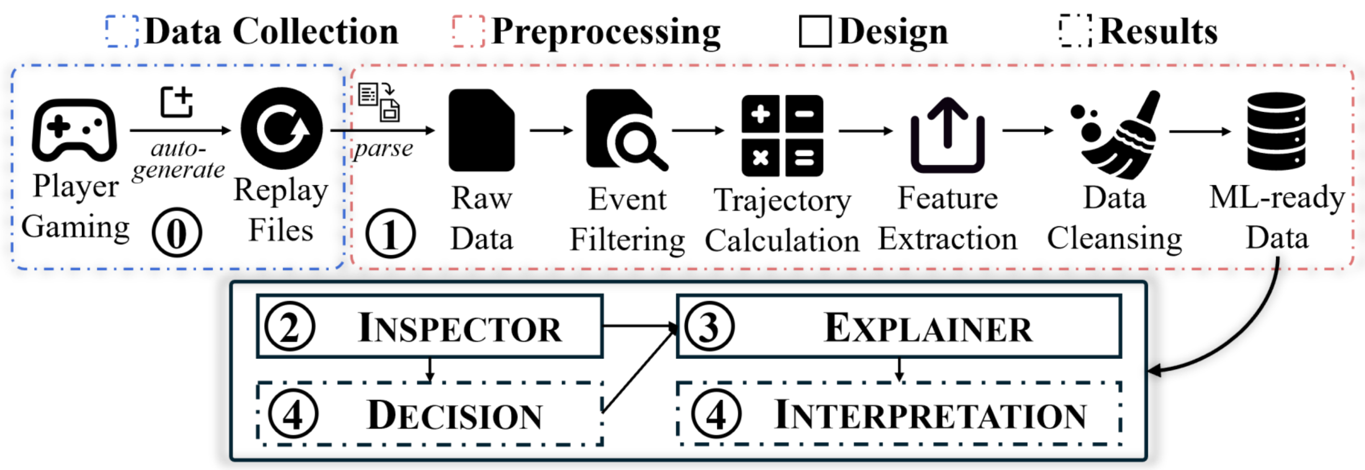

Aimbot


Wallhack

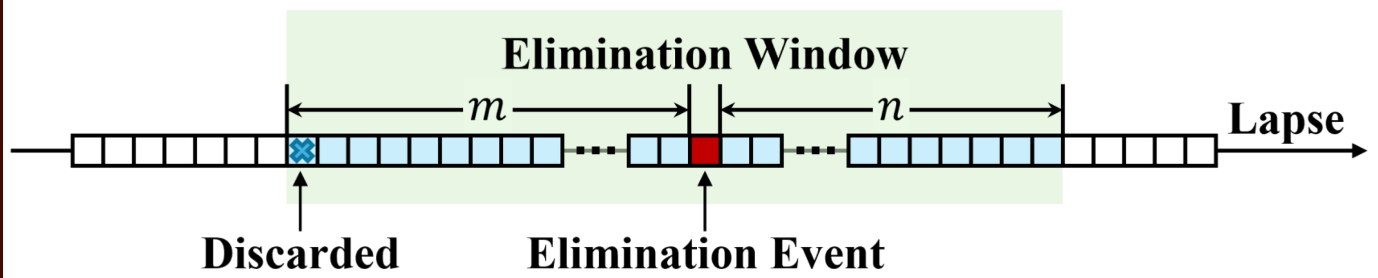
Limitations Addressed

- ✓ Client-side dependency and resource constraints
- ✓ Limited real-world performance
- ✓ Limited generalizability
- ✓ Lack of explainability

Workflow



Trajectory Scope



- ❖ Kill is vital for distinguishing aim-assist cheaters
- ❖ Idle period is the noise for ML model

Links

Artifact

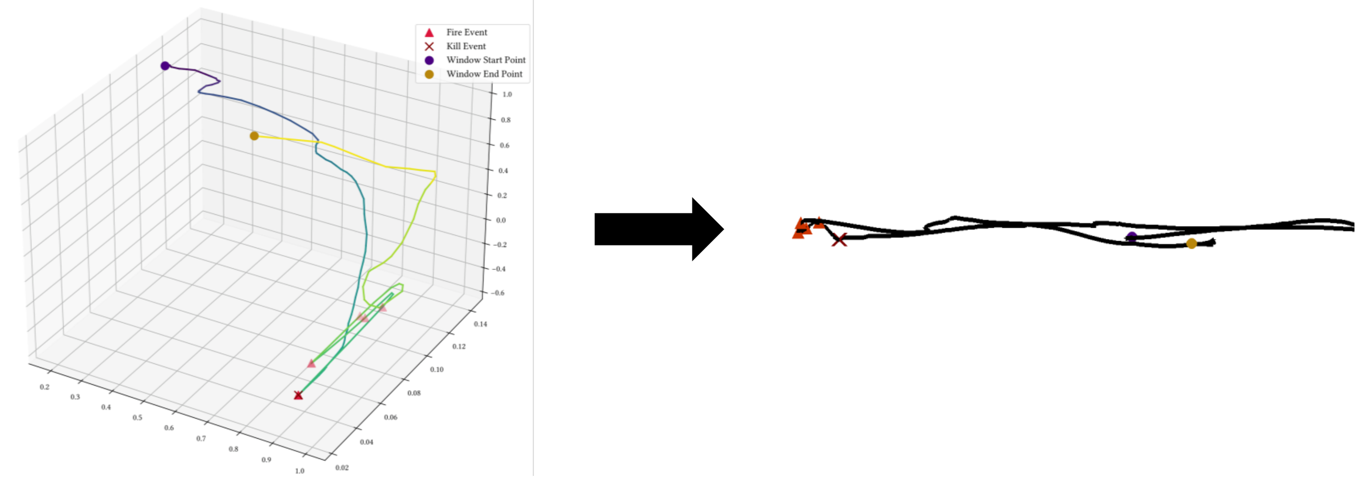
Paper

Demo

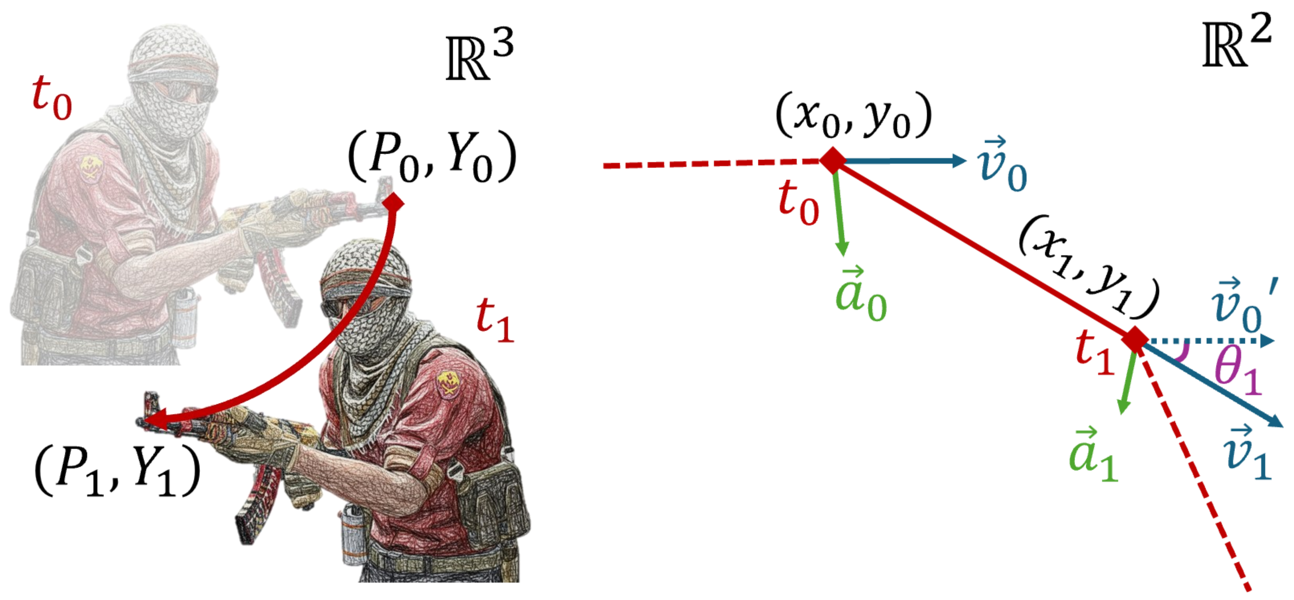
Aim Trajectory Mapping From 3D to 2D

Algorithm 1 Convert Pitch and Yaw to 2D Coordinates

function PITCHYAWTOXY(pitch, yaw, width, height)
  $x \leftarrow \left(\frac{yaw+180}{360}\right) \times width$
  $y \leftarrow \left(\frac{pitch+90}{180}\right) \times height$
  $y \leftarrow height - y$
  return (x, y)
end function



Feature Origination and Construction



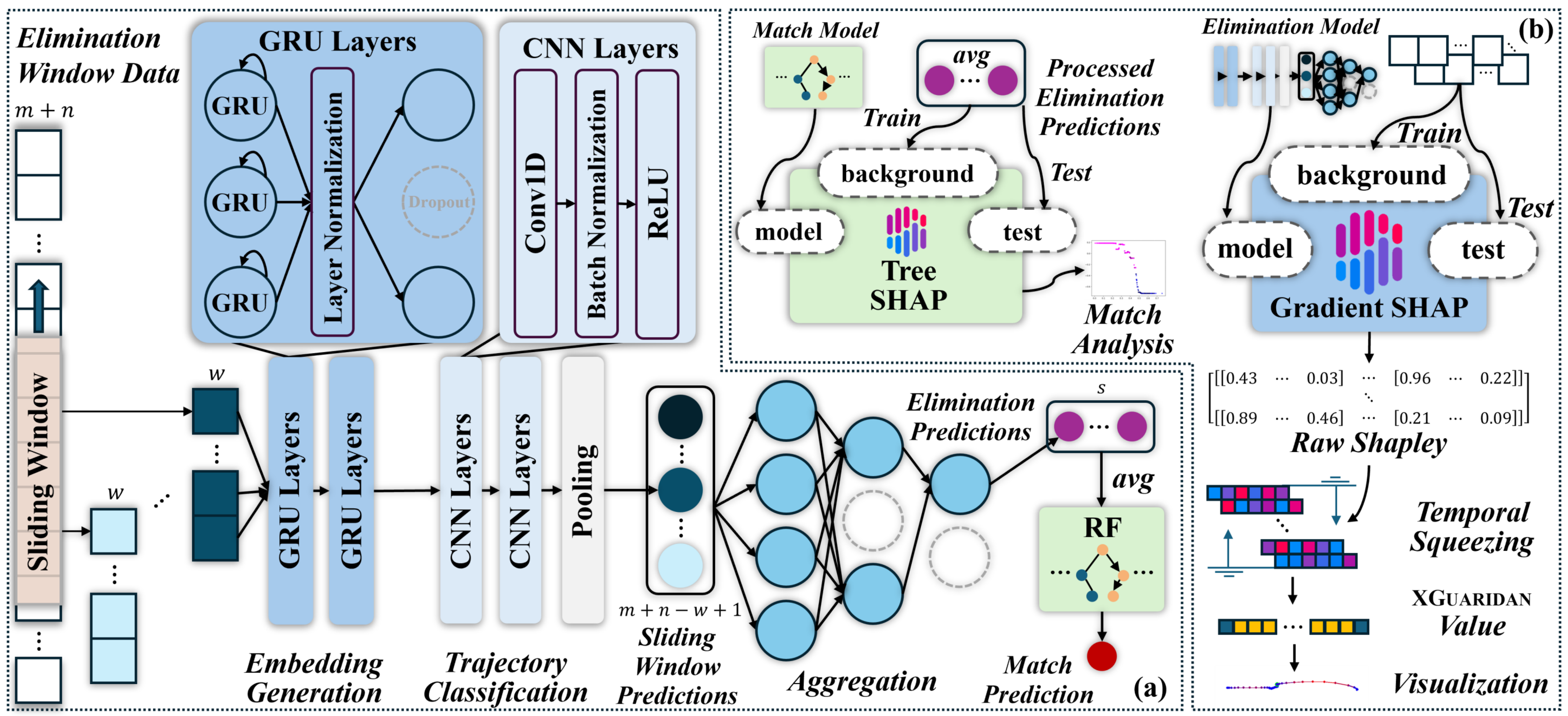
- ❖ Player view orientation in 3D game world: ***pitch***, ***yaw***, and *roll*
- ❖ **Pitch** refers to the vertical angle of the player's view – up-and-down rotation
- ❖ **Yaw** refers to the horizontal angle of the player's view – left-to-right rotation
- ❖ Pitch and yaw are FPS must-haves

For each data point on the trajectory, we calculate

- ❖ **Velocity** ($\vec{v}$) in X and Y direction
- ❖ **Acceleration** ($\vec{a}$) in X and Y direction
- ❖ **Angular change** (θ) of the velocity

$[\boldsymbol{t}, \boldsymbol{I_f}, \boldsymbol{I_e}, \boldsymbol{\vec{v}_x}, \boldsymbol{\vec{v}_y}, \boldsymbol{\vec{a}_x}, \boldsymbol{\vec{a}_y}, \boldsymbol{\theta}]$ **becomes the input**, where t is the absolute tick, I_f and I_e are Boolean values indicating whether the player is firing or killing.

Framework



Dataset

Data Split	A	B	C
#Tick	992,064	1,014,624	1,062,528
#Total Elimination Window	10,334	10,569	11,068
#Cheating Elimination Window	2,385	2,760	2,900
#Normal Elimination Window	7,949	7,809	8,168
#Match	971	944	988
#Cheater	108	115	127
#Normal Player	1,711	1,671	1,754
#Total Player	1,819	1,786	1,881

Key Evaluations

		Accuracy(↑)	FPR(↓)	Recall(↑)	Precision(↑)	F1-Score(↑)
Combination	ABC	0.889	0.062	0.889	0.938	0.907
	ACB	0.876	0.049	0.876	0.951	0.901
	BAC	0.882	0.101	0.882	0.899	0.890
	BCA	0.907	0.041	0.907	0.959	0.924
	CAB	0.872	0.050	0.872	0.950	0.898
	CBA	0.879	0.048	0.879	0.952	0.904
mean (μ)		0.884	0.059	0.884	0.942	0.904
var (σ^2)		1.32E-04	4.00E-04	1.32E-04	4.00E-04	1.08E-04
std (σ)		1.15E-02	2.00E-02	1.15E-02	2.00E-02	1.04E-02

Dataset	Method	Accuracy	FPR	Recall	Precision	F1-Score
XGUARDIAN	TH_ACCA [62]	0.738	0.455	0.738	0.545	0.627
	TH_HITACC [2,22]	0.644	0.086	0.644	0.914	0.711
	XGUARDIAN	**0.889**	**0.062**	**0.889**	**0.938**	**0.907**
HAWK [63]	REVPOV [63]	0.616	0.151	0.616	0.849	0.694
	REVSTATS [63]	0.814	0.109	0.814	0.891	0.842
	EXSPC [63]	0.793	0.114	0.793	0.886	0.827
	HAWK [63]	0.716	0.112	0.716	0.888	0.772
	XGUARDIAN	**0.841**	**0.056**	**0.841**	**0.944**	**0.878**

Dataset	HAWK [63]	FARLIGHT84	XGUARDIAN
Game (Release Year)	CS:GO (2012)	Farlight84 (2023)	CS2 (2023)
Sub-genre	Tactical Shooter	Battle Royal	Tactical Shooter
Engine	Source	Unreal Engine 4	Source 2
Platform	PC	Mobile	PC
Accuracy	0.841	0.816	0.889
FPR	0.056	0.144	0.062
Recall	0.841	0.816	0.889
Precision	0.944	0.856	0.938
F1-Score	0.878	0.829	0.907

Takeaways

- ❖ Provide visual aid to help human reviewers make faster and more accurate decisions.
- ❖ Robust to adversarial attacks when cheaters attempt to mimic normal player behaviors, with only a statistically insignificant drop in recall.
- ❖ Incur minor asynchronous server overhead and only use CPU for prediction, making it practical for real-world deployment.